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Pin tumbler lock

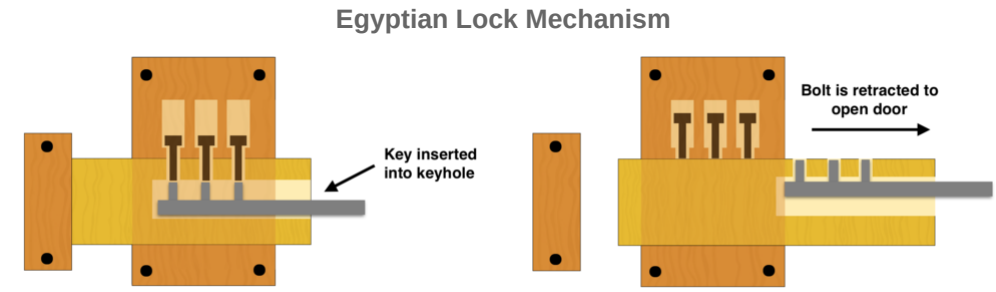
The **pin tumbler lock** is a lock mechanism that uses pins of varying lengths to prevent the lock from opening without the correct key. Pin tumblers are most commonly employed in cylinder locks, but may also be found in tubular pin tumbler locks (also known as radial locks or ace locks).



A common type of pin tumbler lock, of the euro cylinder type.

History

The first known example of a tumbler lock was found in the ruins of the Palace of Khorsabad in Iraq.[1] Basic principles of the pin tumbler lock may date as far back as 2000 BC in Egypt; the lock consisted of a wooden post affixed to the door, and a horizontal bolt that slid into the post. The bolt had vertical openings into which a set of pins fitted. These could be lifted, using a key, to a sufficient height to allow the bolt to move and unlock the door.[2] This wooden lock was one of Egypt's major developments in domestic architecture during classical times.[1]



The key is inserted into the lock bolt, and the pins are lifted

The key is then pulled out of the lock along with the deadbolt

Such a lock, however, may be defeated by lifting the pins uniformly beyond the unlatching point. In 1805, the earliest patent for a double-acting pin tumbler lock — one where lifting the pins too much or too little prevented opening — was granted to American physician Abraham O. Stansbury in England. It was based on earlier Egyptian locks and Joseph Bramah's tubular pin tumbler lock. Two years later, Stansbury was granted a patent in the United States for his lock.[3]

In 1848, Linus Yale Sr. invented the modern pin-tumbler lock.[4] In 1861, Linus Yale Jr., inspired by the original 1840s pin-tumbler lock designed by his father, invented and patented a smaller flat key with serrated edges as well as pins of varying lengths within the lock itself, the same design of the pin-tumbler lock in use today.[5]

Design

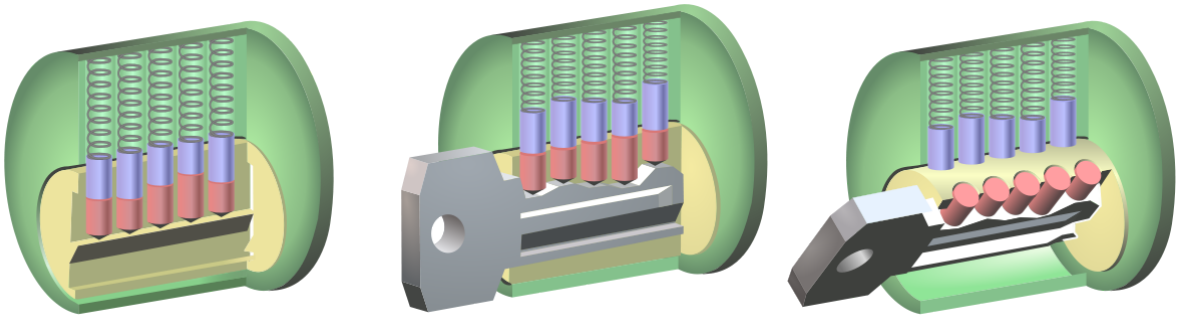
The pin tumbler is commonly used in cylinder locks. In this type of lock, an outer casing has a cylindrical hole in which the *plug* is housed. To open the lock, the plug must rotate.

The plug has a straight-shaped slot known as the *keyway* at one end to allow the key to enter the plug; the other end may have a *cam* or lever, which activates a mechanism to retract a locking bolt. The keyway often has protruding ledges that serve to prevent the key pins from falling into the plug, and to make the lock more resistant to picking. A series of holes, typically five or six of them, are drilled vertically into the plug. These holes contain *key pins* of various lengths, which are rounded to permit the key to slide over them easily.

Above each key pin is a corresponding set of *driver pins*, which are spring-loaded. Simpler locks typically have only one driver pin for each key pin, but locks requiring multi-keyed entry, such as a group of locks having a *master key*, may have extra driver pins known as *spacer pins*. The outer casing has several vertical shafts, which hold the spring-loaded pins.

When the plug and outer casing are assembled, the pins are pushed down into the plug by the springs. The point where the plug and cylinder meet is called the *shear point.* With a key properly cut and inserted into the groove on the end of the plug, the pins will rise causing them to align exactly at the shear point. This allows the plug to rotate, thus opening the lock. When the key is not in the lock, the pins straddle the shear point, preventing the plug from rotating.

Normal Operation

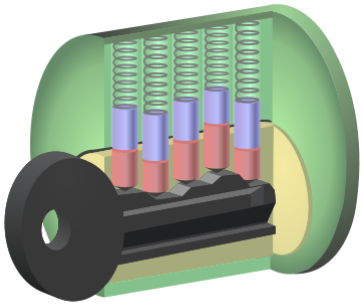


Without a key in the lock, the driver pins (blue) are pushed downwards, preventing the plug (yellow) from rotating.

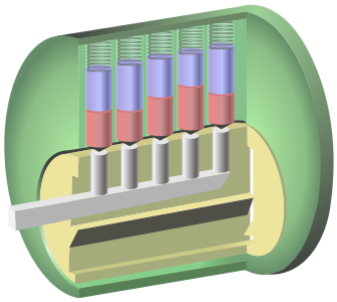
When the correct key is inserted, the gaps between the key pins (red) and driver pins (blue) align with the edge of the plug (yellow).

With the gaps between the pins aligned with the shear line, the plug (yellow) can rotate freely.

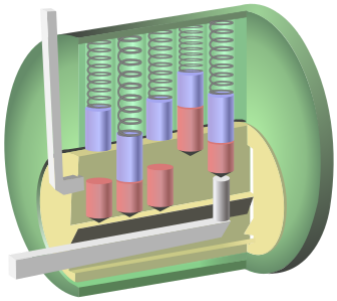
Abnormal Situations



When an incorrect key is inserted into the lock, the key pins (red) and driver pins (blue) do not align with the shear line; therefore, it does not allow the plug (yellow) to rotate.



Some insecure pin tumbler lock models can be opened by pushing all pins above the shearline with a comb pick tool. Some more secure models prevent this attack by limiting the space available within the lock.



A top of the keyway tension wrench has been inserted to apply torque to the lock plug. Pins 1 and 3 have been successfully picked. Pin number 4 has been overset by the picker and will prevent the lock from opening unless it is reset by the picker. Pins 2 is in its natural resting position and remains unpicked. The picker is currently manipulating pin 5 with a lock pick.

Cylinder locks

Commonly pin tumbler locks are found in a cylinder that can be easily unscrewed by a locksmith to facilitate rekeying. The first main advantage to a cylinder lock, also known as a profile cylinder lock or euro, is that the cylinder can be changed without altering the boltwork hardware. Removing the cylinder typically requires only loosening a set screw, then sliding the cylinder from the boltwork. The second is that it is usually possible to obtain, from various lock manufacturers, cylinders in different formats that can all be used with the same type of key. This allows the user to have keyed-alike, and master-keyed systems that incorporate a wide variety of different types of lock, such as nightlatches, deadbolts and roller door locks.

Typically, commercial padlocks can also be included, although these rarely have removable cylinders. Standardised types of cylinder include:

- rim-mounted (also known as *night latch*) cylinders
- Euro cylinders
- key-in-knobset cylinders
- Ingersoll-format cylinders
- American, and Scandinavian round mortise cylinders
- Scandinavian oval cylinders



Euro profile locks, an example of a cylinder lock. These are commonly found on uPVC doors and commercial buildings where re-keying doors is common.

There are also standardised cross-sectional profiles for lock cylinders that may vary in length - for example to suit different door thicknesses. These profiles include the europrofile (or DIN standard), the British oval profile and the Swiss profile

Master keying

Vulnerabilities

See also

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